

Coefficient positivity of stretched Littlewood–Richardson polynomials for partitions of length at most six

Chris Hadad

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Abstract

We prove coefficient positivity at partition length at most six by an exact computer-assisted argument for stretched Littlewood–Richardson polynomials. The argument uses the original hive lattice at every boundary and actual dimension. Its decisive step is a realizability restriction on a complete refined normal-cycle certificate: a normal support whose universal tight-row closure has excessive interior rank cannot carry a face of the required dimension. Its sign constraint may therefore be omitted, while the complete balanced coefficient identity is retained. In rank six this permits an exact rational correction of every contributing edge weight and yields $c_1 \geq (1/2000000) \sum_E \text{vol}_{\mathbb{Z}}(E)$. We assemble the higher-coefficient certificates in the same conventions and give a small independent computation of the top coefficients. The complete finite checks establish strict positivity of every coefficient through the actual degree for a nonempty hive. We separate the geometric proof, exact finite premises, certificate verification, and regeneration costs. The account also explains the scalar-compression and lattice algorithms that make the computation practical, and records the substantial AI contributions and the limits of their independent checks. We additionally prove a constructive legal-realization converse for retained physical branches and give a complete 103-identity presentation of rank-six hive-face closure, certified by short propositional proofs.

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1 Statement and mathematical contribution

Let λ, μ, ν be nonnegative integral partitions with $|\lambda| = |\mu| + |\nu|$. Throughout, λ is the outer partition. For positive integers t , write

$$P_{\lambda, \mu, \nu}(t) = c_{t\mu, t\nu}^{t\lambda}.$$

These values agree with a polynomial in t by stretching polynomiality [8]. The coefficients discussed here are its ordinary monomial coefficients, rather than its values at integers or coefficients in a binomial basis. The King–Tollu–Toumazet conjecture [7] asserts their nonnegativity without a restriction on the partition lengths.

Theorem 1.1 (Rank-six coefficient positivity). *Suppose that the maximum trimmed length of λ, μ, ν is at most six, and let H be their original hive polytope. If H is nonempty and $d = \dim H$, then*

$$P_{\lambda, \mu, \nu}(t) = \sum_{k=0}^d c_k t^k, \quad c_0 = 1, \quad c_k > 0 \quad (1 \leq k \leq d).$$

All coefficients above d vanish. An empty positive-stretch family has the zero polynomial. A point hive, including the all-empty triple, has polynomial one.

The strengthened strict statement uses the small top-coefficient computation in Section 6 together with the lower-coefficient certificates. It is an assembly of analytic theorems and exact finite computations, not a formal proof checked by a proof assistant. The release includes every finite input and verifier identified in Section 7.

For an actual face F , $\text{vol}_{\mathbb{Z}}(F)$ denotes Lebesgue measure normalized by the saturated lattice $\mathbb{Z}^{10} \cap \text{lin}(F - F)$; it is not multiplied by $(\dim F)!$. More precisely, the proof gives

$$c_k(H) \geq \varepsilon_k \sum_{\substack{F \text{ an actual face of } H \\ \dim F = k}} \text{vol}_{\mathbb{Z}}(F), \quad 1 \leq k \leq 9, \tag{1}$$

with

$$\begin{aligned} (\varepsilon_1, \dots, \varepsilon_6) &= \left(\frac{1}{2000000}, \frac{1}{200000000}, \frac{1}{2000000000}, \frac{1}{20000000}, \frac{1}{2000000}, \frac{1}{3000} \right), \\ (\varepsilon_7, \varepsilon_8, \varepsilon_9) &= \left(\frac{1}{144}, \frac{1}{9}, \frac{1}{2} \right). \end{aligned}$$

When $d = 10$, $c_{10} = \text{vol}_{\mathbb{Z}}(H)$. When $d < k$, both the coefficient and the face sum in (1) vanish.

1.1 Why a further step was needed

Ferudun proved positivity through length five [3]. His proof already uses exact normal corrections and the closure relations of two-face polygons. The present work builds on that geometric approach and the Berline–Vergne local Euler–Maclaurin formula [1]. Pointwise nonnegativity of the uncorrected local weights is unavailable: negative local weights occur even on actual hives in the five-part case. Thus positive integer counts, polynomiality, and positivity of the highest coefficients do not by themselves settle the remaining coefficients.

At length six the ambient hive dimension is ten. The linear coefficient involves nine-dimensional normal cones and corrections on eight-normal supports. Direct enumeration of boundary triples is an infinite problem; even a complete bounded census cannot prove the all-size assertion. The finite normal formulation removes that infinity but leaves a substantial exact lattice calculation and a large correction system. Our earlier unpruned correction search did not find a feasible field. This failure was not a proof that its sufficient system was infeasible.

The decisive observation is that many independent normal supports cannot contribute to an actual hive edge. Their tightness forces an additional independent interior equality. The new restriction is justified by an exact geometric lemma; it does not discard a constraint merely because its value is negative. The complete correction identity continues to hold before any sign condition is applied.

1.2 What the proof adds and what it uses

The principal contributions assembled here are the family-dependent realizability criterion, a complete refinement-and-balance argument applicable to hidden affine strata, the exact rank-six correction fields, and practical original-lattice algorithms for their construction and verification. The scalar computation also uses an odd-dimensional reciprocity reduction with a finite-character correction for nontrivial lattice cosets. The strengthened geometry identifies precisely which independent physical branches occur for some legal integral boundary and compatible refinement, at any rank. At rank six, a 103-identity calculus determines every tight-row closure; a small propositional certificate proves its completeness without an extreme-ray completeness oracle. The earlier 102 short identities first fail at a generating set of cardinality six. These are explicit constructions and specializations of classical local Euler–Maclaurin, lattice, reciprocity, and finite Fourier methods; no claim to have originated those classical methods is intended.

The finite-box result for seven parts and outer size at most thirty is a separate theorem. The collection preserves its proof and compares the recent finite-cover preprint of Ghodsi [4]. Neither bounded result is a premise of Theorem 1.1. The theorem here leaves ordinary rank seven and unrestricted KTT positivity open. Novelty comparisons are bounded literature checks, not a claim of exhaustive worldwide priority.

2 Original hive, lattice, and local constants

Put $T_6 = \{(i, j) \in \mathbb{Z}_{\geq 0}^2 : i + j \leq 6\}$, and pad all partitions to six parts. For a hive height function $h : T_6 \rightarrow \mathbb{R}$, use boundary values

$$h(i, 0) = \sum_{a=1}^i \mu_a, \quad h(0, j) = \sum_{a=1}^j \lambda_a, \quad h(6-j, j) = |\mu| + \sum_{a=1}^j \nu_a.$$

The balance identity makes the corner values consistent. The ten free coordinates are the heights at

$$(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (4, 1).$$

Their lattice is $L = \mathbb{Z}^{10}$, with its standard Euclidean metric. For every elementary rhombus the sum at the two obtuse corners minus the sum at the two acute corners is nonnegative. All 45 inequalities are kept. After fixing the boundary they have the form

$$\rho_r(b, x) = \ell_r(b) + \langle n_r, x \rangle \geq 0, \quad 0 \leq r < 45, \quad (2)$$

where $b = (\lambda, \mu, \nu)$. Their interior restrictions give 42 distinct oriented primitive normal directions. Three directions have two physical row preimages. Equal directions do not identify their boundary offsets.

The hive correspondence identifies the integer points of H_b with the LR coefficient [6]; homogeneous boundary scaling gives $H_{tb} = tH_b$ for $t > 0$. Stretching polynomiality implies that its Ehrhart quasipolynomial has period one. The actual degree is $\dim H_b$, which may be smaller than ten. Padding is carried out in this original model; no coordinate upper bound is substituted for the actual dimension.

For a rational hive polytope, choose a positive integer s clearing all its vertices. Its integral dilate satisfies

$$c_k(sH) = s^k c_k(H), \quad \text{vol}_{\mathbb{Z}}(sF) = s^k \text{vol}_{\mathbb{Z}}(F). \quad (3)$$

These identities use period-one polynomiality, not continuity of arbitrary Ehrhart coefficients under perturbation. The integral case has constant term one; hence so does every nonempty period-one rational hive. This also discharges the point and all-empty cases.

Definition 2.1 (Complete local constant). For an independent set I of q original normals, let $\sigma_I = \text{pos}(n_i : i \in I)$. Its constant α_I is the Berline–Vergne dual local value at zero, with the saturated normal lattice $L^* \cap \text{span}(\sigma_I)$ and the induced metric. Equivalently, compute in the projected primal quotient lattice. The normal generators must not be substituted for tangent generators.

For actual integral faces the local Euler–Maclaurin formula expresses c_k as the sum of their normalized volumes times these local values. The precise analytic inputs are [1, Propositions 12–13, Definition 22, Corollary 23, Theorem 26]: quotient and ambient/subspace compatibility, dual solid subdivision additivity, and the local coefficient formula. Section 3 gives the complete argument needed to use simplicial constants without assuming the original polytope is simple or full-dimensional.

3 The complete refined normal cycle

The following construction works in $L = \mathbb{Z}^m$ with a fixed rational positive-definite metric. Let P be a nonempty bounded rational polytope given by finitely many nonzero integral inward normals from a fixed list N . For coefficient statements assume that its Ehrhart counting function is a polynomial, as it is for an integral polytope or an original hive. The geometric refinement and balance lemmas themselves need no polynomiality assumption. Fix $1 \leq k < m$, and put $q = m - k$.

Lemma 3.1 (Compatible refinement using original rows). *The full ambient normal fan of P has a pointed simplicial refinement Γ whose rays come from the original normal directions, including when P has hidden affine equalities.*

Proof. Replace each inequality $b_r + \langle n_r, x \rangle \geq 0$ by $b_r + \langle n_r, x \rangle + \delta a_r \geq 0$, with all $a_r > 0$ generic and $\delta > 0$ sufficiently small. The perturbed polytope is full-dimensional, because any point of P now satisfies every row strictly. It has the same zero recession cone and is bounded. Genericity avoids the finite affine circuit degeneracies, so it is simple.

Every perturbed vertex is given by an independent m -row basis of original inequalities. There are only finitely many such bases. For every basis feasible for arbitrarily small positive δ , its vertex has a finite limit in P , with the same basis tight. That limit is an actual vertex, and its normal cone contains the perturbed vertex cone. A common sufficiently small δ handles every feasible basis. The perturbed normal fan therefore refines the full ambient fan of P . Parallel physical rows retain their separate offsets; genericity selects at most one over a fixed oriented normal on a simple face. No lattice count of the perturbed polytope is used. $\square$

For $\sigma \in \Gamma$ of dimension q , assign weight $w_\sigma = \text{vol}_{\mathbb{Z}}(F)$ when relint σ lies in the normal cone C_F of an actual k -face F , with $\dim C_F = q$, and assign weight zero otherwise. Each top-dimensional subdivision cell of C_F receives the same full actual face volume.

For a $(q-1)$ -cone τ , let $u_{\sigma/\tau}$ denote the primitive generator of the image of σ in $L^*/(L^* \cap \text{span } \tau)$.

Lemma 3.2 (Complete primitive balance). *For every $(q-1)$ -cone τ of Γ ,*

$$\sum_{\substack{\sigma \supset \tau \\ \dim \sigma = q}} w_\sigma u_{\sigma/\tau} = 0. \quad (4)$$

Proof. If relint τ lies in a coarse cone of dimension $q-1$, that cone corresponds to an actual $(k+1)$ -face G . Each adjacent nonzero-weight cell corresponds to one facet F of G near relint τ . Put $M_G = L \cap \text{lin}(G - G)$. Restriction $L^* \rightarrow M_G^*$ is surjective because M_G is saturated; its kernel is $L^* \cap \text{span } \tau$. For the primitive facet conormal u , $\text{covol}(M_F) = \text{covol}(M_G) \|u\|$. Indeed, extend a basis of its kernel by a lattice vector with $u(v) = 1$; the extra perpendicular height is $1/\|u\|$. The Euclidean facet balance formula, divided by the common covolume, is exactly (4). An overall reversal from outward to inward normals does not change the zero sum.

If relint τ lies inside a coarse q -cone, exactly two adjacent q -cells in its span contribute. Their weights agree and their primitive quotient directions are opposite. If it lies inside a coarse cone of larger dimension, all adjacent q -cell weights are zero. These cases exhaust the refinement, including coarse cones with lineality. $\square$

For every independent $(q-1)$ -support J , choose a rational linear functional h_J on its quotient lattice, allowing the zero functional. For $I = J \cup \{i\}$, choose a saturated integer kernel basis M_J of the original normal matrix. In its coordinates,

$$u_{I/J} = \frac{M_J^T n_i}{\text{gcd of all coordinates of } M_J^T n_i}. \quad (5)$$

Independence makes the denominator positive. Define

$$\beta_I = \alpha_I + \sum_{J \text{ a facet support of } I} h_J(u_{I/J}). \quad (6)$$

Proposition 3.3 (Correction identity). *For every such rational field h ,*

$$c_k(P) = \sum_{\substack{\sigma \in \Gamma \\ \dim \sigma = q}} w_\sigma \alpha_\sigma = \sum_{\substack{\sigma \in \Gamma \\ \dim \sigma = q}} w_\sigma \beta_\sigma. \quad (7)$$

Proof. First dilate to make P integral. On each actual normal cone, BV dual solid valuation adds the local constants of all top-dimensional simplicial cells; lower-dimensional intersections contribute zero in that fixed span. For a nonpointed coarse cone, ambient/subspace compatibility identifies the value with the transverse value in the actual affine hull. The local Euler–Maclaurin formula gives the first equality. Insert (6) and interchange the finite sums. The coefficient of every functional h_J is the complete sum (4), so all corrections cancel. Transfer back using (3). $\square$

Corollary 3.4. *If $\beta_I \geq \varepsilon > 0$ on every independent q -support, then $c_k(P) \geq \varepsilon \sum_{\dim F=k} \text{vol}_{\mathbb{Z}}(F)$.*

Proof. All weights are nonnegative. Every actual k -face has at least one top-dimensional normal subdivision cell. Sum (7); use (3) for rational P . $\square$

This is a sufficient field criterion. Failure to find such a field does not exhibit a polytope, much less a whole LR polynomial, with a negative coefficient. Abstract nonnegative balanced weights need not be realizable face-volume weights.

4 Realizability pruning

Consider a homogeneous polyhedral family $P_b = \{x : \rho_r(b, x) \geq 0\}$, where each ρ_r is linear in the combined boundary and interior coordinates. Let C be the cone of all (b, x) satisfying the physical inequalities; a larger cone containing every permitted family member is also safe. For a physical row set S , set

$$C_S = C \cap \{\rho_r = 0 : r \in S\}, \quad \text{cl}(S) = \{r : \rho_r \text{ vanishes identically on } C_S\}.$$

Given independent normal directions I , a *physical branch* chooses one original row above each direction. Retain I if at least one such branch S has

$$\text{rank}\{n_r : r \in \text{cl}(S)\} = |I|. \quad (8)$$

For a specified boundary this is a necessary support condition. The converse below is existential: every retained physical branch occurs at some legal integral boundary in some compatible refinement. It does not specify the same boundary or refinement for different branches.

Lemma 4.1 (Support of positive weights). *Every q -cell of the refinement in Lemma 3.1 with positive weight has a retained physical branch.*

Proof. Let σ contribute the volume of an actual k -face F . Choose its q defining physical rows in the relaxed polytope, and an incident relaxed vertex whose independent m -row basis contains them. Its limit is an actual vertex v of P_b , with these rows tight. For $\xi \in \text{relint } \sigma \subset \text{relint } C_F$, minimization at v shows $v \in F$. Each selected normal belongs to C_F , so its row is constant on F and, being zero at v , is tight on all of F .

Let $T(F)$ consist of all original rows tight on F , including every hidden affine equality. Then $S \subset T(F)$ and $\text{rank}_x T(F) = m - \dim F = q$. A relative-interior point (b, x) of F lies in C_S . Every row in $\text{cl}(S)$ vanishes there. Its restriction to F is a nonnegative affine function, so vanishing at a relative-interior point forces it to vanish identically. Therefore

$$S \subset \text{cl}(S) \subset T(F), \quad q \leq \text{rank}_x \text{cl}(S) \leq q.$$

This proves (8) for the actual branch. Taking an existential choice over all branches cannot discard it. $\square$

Theorem 4.2 (Family-dependent sign certificate). *If the field in (6) has $\beta_I \geq \varepsilon > 0$ for every retained independent q -support, then every nonempty period-one member of the family satisfies*

$$c_k(P_b) \geq \varepsilon \sum_{\dim F=k} \text{vol}_{\mathbb{Z}}(F).$$

Proof. Proposition 3.3 holds for every field, regardless of sign. Lemma 4.1 puts every positive-weight cell in the retained set. Excluded cells have weight zero and impose no sign condition. Corollary 3.4's final summation and rational scaling now apply without altering any incidence or balance identity. $\square$

For rank six, the universal cone uses 28 heights and all 45 original rhombi. Its affine lineality has dimension three. Set $h(0,0) = h(1,0) = h(0,1) = 0$; subtracting an affine height function gives this gauge and leaves every rhombus slack unchanged. The resulting cone is pointed and full-dimensional in 25 coordinates. Its 166 primitive extreme rays provide an exact closure test: intersect the ray sets on which every selected physical row is tight, and find every row vanishing on all the surviving rays. If there is no surviving ray, the face is the origin and every row belongs to its closure. Cone completeness must be verified; an incomplete ray list could make this exclusion unsound. Section 4.2 gives an alternative complete closure certificate from feasible vectors and positive identities. Their 166 tight masks agree exactly with the original classifier, and all six physical-row symmetries preserve this set of masks. Either complete route certifies the closure predicate used below.

For nine independent normals, compute an exact nonzero kernel vector $w \in \mathbb{Q}^{10}$. Condition (8) is equivalent to every closure row being orthogonal to w . All branches over the three repeated directions are checked. The six barycentric coordinate permutations preserve physical rows, the cone, the original lattice and its metric. Thus the predicate is invariant under that six-element group.

4.1 Exact legal realization of retained branches

The support criterion has a constructive geometric converse, valid beyond the rank-six certificate. Put $m = (n-1)(n-2)/2$ and $T_n = \{(i,j) \in \mathbb{Z}_{\geq 0}^2 : i+j \leq n\}$, retaining every physical rhombus on T_n .

Theorem 4.3 (Legal realization). *Let $n \geq 3$, and let S contain q physical rows with independent interior normals, where $1 \leq q \leq m$. Then $\text{rank}_x \text{cl}(S) = q$ if and only if there are a legal integral LR boundary and a compatible positive generic original-row relaxation in which the selected physical branch S defines a q -dimensional normal cell of positive actual $(m-q)$ -face weight. The parent may be lower-dimensional or nonsimple. The face uses the original saturated integer lattice.*

Proof. Use the boundary convention of Section 2. For arbitrary n , the three types of rhombi, where all four points exist, are

$$\begin{aligned} A_{i,j} &= -h(i,j) + h(i+1,j) + h(i,j+1) - h(i+1,j+1), \\ B_{i,j} &= -h(i,j) + h(i,j+1) + h(i-1,j+1) - h(i-1,j+2), \\ C_{i,j} &= -h(i,j) + h(i-1,j+1) + h(i-1,j) - h(i-2,j+1). \end{aligned}$$

Define the three boundary words by successive differences along the same horizontal, vertical and

diagonal edges. Their trace is $|\lambda| = |\mu| + |\nu|$. Direct cancellation gives

$$\begin{aligned}\mu_i - \mu_{i+1} &= A_{i-1,0} + C_{i+1,0}, \\ \lambda_j - \lambda_{j+1} &= A_{0,j-1} + B_{1,j-1}, \\ \nu_j - \nu_{j+1} &= B_{n-j,j-1} + C_{n-j+1,j-1}.\end{aligned}$$

All words are therefore weakly decreasing. For integral heights set

$$u = \max(0, -\mu_n), \quad v = \max(0, -\lambda_n, u - \nu_n).$$

Adding $ui + vj - h(0, 0)$ to $h(i, j)$ changes the words to $(\lambda + v, \mu + u, \nu + v - u)$, which are nonnegative integral partitions. Every rhombus slack and the trace are unchanged. The fixed-boundary interior lattice is translated by an integer vector.

Choose a rational relative-interior point of the universal face defined by S , clear its denominators by a positive integer scaling, and apply this legalization. Write it as (b, x_*) , and set $T = \text{cl}(S)$. Exactly the rows T are tight. In the fixed-boundary hive, the face cut out by S has direction space $\ker_{\mathbb{R}} N_T$: every T -row vanishes on it, while sufficiently small displacements in either direction of that kernel preserve every other strict inequality. Its dimension is $m - q$, with affine lattice $x_* + \ker_{\mathbb{Z}} N_T$. This kernel is saturated, directly from the integer matrix equations.

The hive is bounded. A recession direction d has zero boundary. Summing $A_{a,b}(d)$ over $0 \leq a < i$, $0 \leq b < j$ gives $-d(i, j)$, hence $d(i, j) \leq 0$. Also

$$A_{i,j-1}(d) + B_{i+1,j-1}(d) = 2d(i, j) - d(i, j-1) - d(i, j+1) \geq 0.$$

Each vertical sequence is discretely concave with zero endpoints, so $d(i, j) \geq 0$. The recession cone is therefore zero.

Take a rational right inverse Y of N_S , put $y_0 = -Y\mathbf{1}$, and initially let $a_s = 1$ on S and $a_r = \max(1, 1 - n_r y_0)$ otherwise. Multiply y_0 and all a_r by a common positive denominator, leaving Y unchanged. They are now integral, with selected margins zero and off- S margins at least one. Choose an integer M_0 and define

$$B = 2^m, \quad C_* = \max_r \sum_{s \in S} |(n_r Y)_s|, \quad M_0 > \max\{4B + 4, 2(C_* + 1)\}, \quad \tau = M_0^{-1}.$$

In the fixed zero-based physical row order put

$$d_r = a_r + \tau^{r+1}, \quad y = y_0 - Y(\tau^{s+1})_{s \in S}.$$

Every $d_r > 0$; on S , $n_s y + d_s = 0$, and off S , $n_r y + d_r > 1/2$. Thus $x_* + \eta y$ has exactly S tight in $P_b(\eta) = \{x : \rho_r(b, x) + \eta d_r \geq 0\}$.

Each primitive integral circuit of the interior normals has coefficients of magnitude at most B : use signed maximal minors and Hadamard's inequality, since every normal has Euclidean norm at most two. For such a circuit c , $c \cdot a$ is an integer. If nonzero, the perturbation has magnitude at most $B\tau/(1 - \tau) < 1/2$ and cannot cancel it. If zero, the first nonzero power of τ dominates all later terms by the same geometric-series estimate. Hence $c \cdot d \neq 0$, including circuits of parallel physical rows.

An explicit compatible amplitude is any rational

$$0 < \eta < \frac{1}{8mB^2A_*}, \quad A_* = \max_r a_r + 1.$$

The relaxed polytope is nonempty, full-dimensional and bounded. For an independent m -row basis, an originally negative slack has magnitude at least $1/B$, since its denominator divides the basis determinant. Cofactors have magnitude at most 2^{m-1} ; a normal has at most four nonzero entries. The change in any slack is at most $\eta(2mB + 1)A_* < 1/(2B)$. Thus no infeasible limiting basis becomes feasible. A dependent active set would contain a circuit satisfying $c \cdot \ell(b) + \eta c \cdot d = 0$. The first term is integral; genericity handles zero, and $\eta|c \cdot d| < 1$ handles a nonzero value. The relaxation is simple, and every vertex cone lies in the normal cone of its original limiting vertex. Its entire fan is compatible.

For $\xi = \sum_{s \in S} t_s n_s$, with every $t_s > 0$, the original minimizers are exactly the face defined by S : a positive sum of its nonnegative slacks vanishes precisely there. The selected normal cell thus has the positive saturated volume of that actual face as its weight, with zero-dimensional volume one. Necessity is the tight-row rank argument of Lemma 4.1, which also applies to zero-dimensional faces. $\square$

This identifies the existential geometric support of the criterion. It does not make every retained field inequality individually necessary for positivity: different fields, compatible refinements and cancellations remain available.

4.2 A short calculus for every rank-six closure

There is a compact certificate for the closure computation itself. Its zero-based physical row order agrees with the original atlas.

Theorem 4.4 (Positive closure calculus). *Among the 45 rank-six rhombi, take all disjoint unit-coefficient identities with two rows on each side and with three rows on each side. There are 30 and 72, respectively, counting reversal as the same identity. Add*

$$\begin{aligned} C_{2,1} + B_{1,2} + A_{1,3} + A_{2,0} + C_{4,2} + B_{4,0} &= A_{0,2} + C_{2,3} + B_{2,0} \\ &\quad + C_{4,0} + A_{3,1} + B_{3,2}. \end{aligned}$$

These 103 positive identities generate exact physical-row closure for every subset of the 45 rows. The original 102-identity calculus first fails at generating-set cardinality six.

An identity gives two implications: if all rows on one positive side vanish, every row on the other side vanishes. The 206 set implications produce 564 distinct single-conclusion Horn clauses. Iteration adds at most 45 rows. The displayed identity is a direct equality of all 28 height coefficients; the individual equality is not asserted new in the literature.

Proof from the small exact certificate. Let $R(S)$ be rule closure, $C(S)$ actual cone closure, and $V(S)$ closure on the 166 supplied feasible integral heights. Sound positive identities and feasible heights give

$$R(S) \subseteq C(S) \subseteq V(S).$$

For each target r , introduce Boolean variables for membership in a set M . Include $\neg x_r$, all Horn implications, and for every supplied height where $\rho_r > 0$, the clause $\bigvee_{i:\rho_i>0} x_i$. A satisfying assignment is exactly a rule-closed M with $r \in V(M) \setminus M$.

The certificate supplies nine representative reverse-unit-propagation refutations, with 6,488 additions, and 45 exact variable transports of their complete input clause sets. For an added clause, negate every literal and unit-propagate the existing database; a contradiction proves that clause is a consequence. Every representative ends in a checked empty clause. Exact clause-set equality

under a variable bijection transports unsatisfiability. Hence $V(M) = M$ for every rule-closed set. With $M = R(S)$, monotonicity gives $C(S) \subseteq C(M) = M$, proving $C(S) = R(S)$.

The independent checker reconstructs every literal rhombus and enumerates every short identity. It checks 2,884 identity coordinates and 7,470 feasibility slacks, regenerates each input formula, checks every proof addition and verifies every target transport. It uses no SAT solver or extreme-ray completeness oracle. The 254,843-byte certificate takes about five seconds to check with the independent Python standard-library program.

For minimality, the old-rule-closed set $S = \{4, 17, 18, 23, 31, 41\}$ has actual closure

$$\{2, 3, 4, 7, 8, 9, 11, 17, 18, 23, 24, 26, 31, 35, 39, 40, 41, 44\}.$$

Its interior rank rises from six to eight. The new identity and old rules force these rows; the sum of the feasible heights tight on S has exactly this zero set, proving equality. Separately, all seeds of size zero through five are checked. Their populations are 1, 45, 990, 14190, 148995, 1221759, totaling 1,385,980; each has equal old-rule and feasible-witness closures, which now equal actual closure. This proves minimum cardinality six, without uniqueness or a minimum-size presentation claim. $\square$

The sandwich argument also shows that summing feasible heights tight on any seed gives an integral relative-interior witness for its exact closure. Together with Theorem 4.3, this gives a constructive physical-branch interface. The large rational positivity field and its local-scalar obligations remain separate.

5 Exact scalar computation and its acceleration

Let N be any independent q -row normal matrix. In normal-image coordinates use

$$\Lambda = N\mathbb{Z}^{10}, \quad I = [\mathbb{Z}^q : \Lambda], \quad G = NN^T, \quad Q = G^{-1}.$$

Let d_i be the least positive integer with $d_i e_i \in \Lambda$, and $\mathcal{F} = \Lambda \cap \prod_i [0, d_i)$. Unique reduction modulo the primitive axes gives

$$S(c) = \frac{\sum_{p \in \mathcal{F}} e^{\langle c, p \rangle}}{\prod_i (1 - e^{d_i c_i})}, \quad |\mathcal{F}| = \frac{\prod_i d_i}{I}. \quad (9)$$

Every fundamental numerator point is needed. Equal Gram matrices and equal indices alone need not identify the full image lattice. Type reuse must verify its original columns and exact row/ambient transport.

Let R_Q be the degree-preserving holomorphic projection for metric Q . The local constant is $\alpha_N = R_Q(S^{[0]})$, where $[0]$ means total homogeneous degree zero [2, 5]. This projection precedes any one-variable specialization. A Laurent constant on a generic ray is a different operation. With $B_a(x)$ the Bernoulli polynomial in $ze^{xz}/(e^z - 1) = \sum_{a \geq 0} B_a(x) z^a / a!$, put

$$W(a) = \sum_{p \in \mathcal{F}} \prod_i \frac{d_i^{a_i} B_{a_i}(p_i/d_i)}{a_i!}.$$

Then the following finite formula specifies every scalar without a cache:

$$\alpha_N = \frac{(-1)^q}{\prod_i d_i} \sum_{\substack{a \in \mathbb{Z}_{\geq 0}^q \\ |a|=q}} W(a) T(\{i : a_i = 0\}, (\max(a_i - 1, 0))_i). \quad (10)$$

Here $T(\emptyset, 0) = 1$. If $j \notin U$, $e_j > 0$ and $|e| = |U|$, set $b = Q_{UU}^{-1} Q_{Uj}$ and use

$$T(U, e) = \sum_{i \in U} b_i T(U \setminus \{i\}, e - \mathbf{e}_i). \quad (11)$$

To prove it, split c_j into its projection on the denominator span and its orthogonal remainder. In every remainder term, expand the other factors into the same two spaces. The denominator-space factor has negative homogeneous degree, whose holomorphic projection is zero. Orthogonal multiplicativity kills that term. The projected part cancels one denominator and yields (11). Induction finishes the proof.

An independent implementation uses the full Euler–Maclaurin recurrence. For T a row subset, write $I_T = [\mathbb{Z}^T : N_T \mathbb{Z}^{10}]$ and $G_T = N_T N_T^T$, with $I_\emptyset = \alpha_\emptyset = 1$. Choose a common generic vector w , and set $c_T = G_T^{-1} w_T$. With $\mu_\emptyset = 1$, the recurrence is

$$\mu_T(s) = S_T(s c_T) - \sum_{\emptyset \neq U \subseteq T} \frac{(-1)^{|U|} I_{T \setminus U}}{I_T s^{|U|} \prod_{i \in U} c_{T,i}} \mu_{T \setminus U}(s). \quad (12)$$

The quotient covectors are compatible restrictions of the same specialization; all negative Laurent coefficients cancel. Multiplying by $s^{|T|}$ reduces scalar extraction to finite polynomial arithmetic. Higher Taylor coefficients needed by larger supports are retained to the required degree.

5.1 Odd dimension and exceptional lattice cosets

For odd q , proper-subset constants can replace a full degree- q jet calculation. Put $\ell(c) = \sum_i c_i$, and for $F \subseteq T$ define

$$K(\emptyset) = 1, \quad a(F) = Q_{FF}^{-1} (Q \mathbf{1})_F, \quad K(F) = \sum_{i \in F} a_i(F) K(F \setminus \{i\}).$$

Nested orthogonal projection proves the last recurrence by cancelling one numerator factor. If $\mathbf{1} \in \Lambda$, closed/open reciprocity gives $S(-c) = -e^{\ell(c)} S(c)$. Apply R_Q , insert the complete local recurrence, and compare degree zero. A proper germ of positive degree is orthogonal to the denominator variables and has zero contribution; the surviving expression is

$$A_T = -\frac{1}{2} \sum_{J \subsetneq T} (-1)^{|T \setminus J|} \frac{I_J}{I_T} \alpha_J \frac{K(T \setminus J)}{|T \setminus J|!}, \quad \alpha_T = A_T. \quad (13)$$

Thus at $q = 9$ there are 511 proper terms, and only lower constants are required. Original lattice membership is essential.

For general Λ , let S_a be the closed orthant sum on $a + \Lambda$, and $D = S_0 - S_{-1}$. The two coset reciprocity identities give $D(-c) = e^{\ell(c)} D(c)$, so instead

$$\alpha_T = A_T + \frac{1}{2} R_Q(D^{[0]}). \quad (14)$$

Let $\widehat{H}$ be the character group of $H = \mathbb{Z}^q / \Lambda$. For $z_i = \chi(e_i)$, character orthogonality makes the defect explicit:

$$D(c) = \frac{1}{I} \sum_{\chi \in \widehat{H}} \left(1 - \prod_i z_i \right) \prod_i (1 - z_i e^{c_i})^{-1}. \quad (15)$$

All characters, their actual axis phases and the factor $1/I$ are retained. Terms with $\prod_i z_i = 1$ vanish. The other terms are evaluated exactly in a cyclotomic field, or by complete rational Galois traces followed by (11).

For independent primitive rows with entries in $\{0, 1, -1\}$, a nontrivial character cannot have only one nontrivial coordinate. If it has exactly two, column annihilation forces identical row supports and phases equal or opposite. Independence rules out a single relative sign; both signs force order two, with both phases -1 . Their product is one, so the defect vanishes. Every surviving character therefore has at least three nontrivial phases. Its coordinate poles number at most $q - 3$, which is six at $q = 9$. This is the scalar acceleration used in the large certificate; it does not assert positivity of the raw scalars.

6 Finite certificates and assembly

All finite tests use exact integers and rationals. Floating optimization proposes a field only; it contributes no sign decision to the proof. The six coordinate isometries act on complete original supports. Averaging a field seeded at support-orbit representatives includes every stabilizer:

$$\beta_I = \alpha_I + \frac{B_I n}{6D}. \quad (16)$$

Formula (16) is the orbit-averaged convention for the $c_1, \dots, c_5$ fields. The c_6 certificate instead lists its 381 literal support functionals directly, with denominator 10^{12} , and uses (6) without an additional averaging factor. At each support, the kernel matrix M has an integer left inverse A satisfying $NM = 0$ and $AM = I$, and the expected rational rank, or equivalently verifies full rank and gcd one of its maximal minors. This proves that its column lattice is the full saturated kernel. Every restricted conormal is divided by the gcd of all its coordinates. The complete original-normal action is checked against any stored sparse matrix action; agreement of only floating residuals is insufficient.

Coefficient	Normal size	Finite sign domain	Bound
c_1	9	13,325,662 retained orbit rows	1/2000000
c_2	8	10,480,218 orbit rows	1/200000000
c_3	7	2,999,563 orbit rows	1/200000000
c_4	6	675,721 orbit rows	1/2000000
c_5	5	121,241 orbit rows	1/2000000
c_6	4	102,297 literal independent supports	1/3000
c_7	3	10,994 literal independent supports	1/144
c_8	2	849 literal independent supports	1/9
c_9	1	42 literal supports	1/2

The different rows explicitly distinguish literal supports from symmetry representatives. All dependent complements and all transports belong to their coverage checks.

6.1 The linear certificate

The complete nine-normal roster has 27,230,728 orbit rows. The independent closure test visits 46,327,714 physical branches, retains 13,325,662 rows representing 79,940,357 original parents, and excludes the remaining rows from the sign test by Lemma 4.1. It does not delete them from the

balanced identity. The integer field has 20,960,436 coordinates and $D = 10^{10}$ in (16); its actual denominator is $6 \cdot 10^{10}$. The complete exact retained minimum is

$$\frac{4014288690844119591010700379369}{40149066134252712057708583296000000000} > \frac{1}{2000000}.$$

It occurs at zero-based row 8,203,248. The accepted original-action check reconstructs 119,963,492 primitive incidences. The 2,695,883 negative excluded rows remain useful adverse controls, not LR counterexamples.

The full roster is generated from every independent eight-support orbit: extend by each absent normal, test independence in its saturated two-coordinate kernel, and canonicalize under all six actions. Conversely, every independent nine-support has an independent deleted eight-face, so some extension produces its orbit. The verifier checks the complete eight-domain, every extension, duplicate identity, deleted face, multiplicity and source join. A row count by itself would not prove coverage.

6.2 The higher certificates and a small strengthening

For $c_2, \dots, c_6$, the rational fields constrain every independent support, without the family-dependent pruning. Their complete domains, original lattices, primitive incidences, scalar values and corrected inequalities are specified in the corresponding verification modules. Corollary 3.4 applies in every actual dimension.

For $q = 1, 2, 3$, no correction is needed. Reconstructing the 45 rhombi and 42 directions and enumerating all $\binom{42}{q}$ subsets gives 42, 849, 10,994 independent supports and 0, 12, 486 dependent supports. The exact minima are $1/2, 1/9, 1/144$. An independent Python rational implementation of (10) agrees on every support and lattice index with the C++ implementation of (12). Complete nonunimodular numerators are included. This small rank-six calculation replaces a dependency on a rank-uniform top-coefficient catalog; it makes no new rank-uniform assertion.

Proof of Theorem 1.1 from the finite premises. The linear certificate and Theorem 4.2 establish (1) for $k = 1$. The complete higher fields and Corollary 3.4 establish it for $2 \leq k \leq 6$. The small uncorrected certificates establish it for $7 \leq k \leq 9$. Every nonempty polytope of dimension at least k has an actual k -face of positive normalized volume. The leading ten-dimensional coefficient is its volume. Thus all coefficients through actual degree are positive. Section 2 supplies constant one, actual degree, padding, rational scaling and the empty/point cases. No generic-boundary assumption or limiting transfer across chambers is needed. $\square$

7 Verification contract and reproducibility

A certificate is the rational field together with the exact finite domain and the mathematical interpretation above. A file hash establishes identity; it does not prove a scalar value or a domain complete. The distribution must therefore distinguish the following operations.

1. Reconstruct the original rhombi, lattice and symmetry; verify the universal cone and all physical-branch closure tests.
2. Regenerate all required normal supports, saturated kernels, image lattices, type transports and primitive incidences. Reject omissions, duplicates, foreign rows, and incomplete tails.

3. Reconstruct every local scalar from the stated analytic recurrence, including its complete numerator and exceptional lattice cosets. A proper-subset cache must itself have a complete analytic verification.
4. Evaluate every corrected value exactly and compare it with its declared bound. Reconstruct the original action independently of any stored sparse matrix. Record exact minima and complete domain counts.
5. Verify corrupt-data and omission controls, process exit, and complete output before accepting a pass. Reproduce from the distributed source/data copy, not from paths in the private research workspace.

The accompanying package is version *rank-six-2026-09-25-v1* in the [public source repository and release archives](#). Its [rank-six verification guide](#) gives the exact commands, source and data manifests, complete coverage reports, measured costs and negative controls. The proof and the small geometric certificate can be inspected before downloading the large numerical assets.

The fast route rechecks geometry, original action and inequalities using supplied exact scalar data. Full regeneration additionally establishes those scalar inputs and their lower-type dependencies. These are different trust boundaries. The universal cone has a small independent reconstruction: choose 25 independent original inequalities after fixing the affine-height gauge, start with their simplicial cone, and intersect the remaining 20 halfspaces by exact double description. The standard-library Python implementation checks all extreme-ray and facet ranks and gives precisely the same 166 rays and tight masks as a separate Sage/PPL reconstruction. It takes approximately three seconds locally and removes Sage as a required reader dependency.

The fresh public c_1 geometry, classification, original-action and coverage checks took 21.49 minutes on the publication host under concurrent load. The complete independent $q = 9$ scalar check covered all 27,230,728 rows in 3.49 elapsed hours with up to six children; the sum of their elapsed timers was 20.66 hours. These are wall and aggregate child durations, not measured CPU-hours or performance promises. Complete lower scalar regeneration is a separate requirement, recorded with its commands and costs in the verification report. The small top-coefficient replay, including compilation and both implementations, took approximately six seconds locally.

The code uses arbitrary-precision arithmetic where the proof needs it. Machine-width formats are accompanied by exact range and overflow checks. The mathematical trust base comprises the cited classical theorems, the proof in Sections 3–4, and the specified finite predicates. The execution trust base additionally includes the compiler, integer/rational libraries, operating system and hardware. Independent implementations reduce shared implementation risk but do not replace an audit of the common mathematical specification.

7.1 Formal verification

No Lean, Coq or HOL verification of the whole theorem is claimed. A useful formalization must connect the concrete LR coefficient to the original hive count, prove the polynomial/coefficient interpretation, formalize the saturated-lattice normal-cycle and pruning arguments, and certify the exact finite computation. Proving only that a nonnegative weighted sum is nonnegative while assuming this entire connection would leave the central obligation untouched. The accompanying formalization assessment separates these obligations and identifies the exact certificate predicates suitable for a small kernel-checked arithmetic layer. The new closure certificate offers a concrete first target: Lean’s standard library has a sound LRAT checker to which the RUP additions can be exported with ordered unit-propagation hints. That route has been investigated but not compiled

or kernel-checked here. A successful finite CNF check would still require the formula interpretation, closure sandwich and remaining LR/BV bridge to be formalized.

8 Mathematical and computational implications

Realizability pruning is available for any homogeneous fixed-row polyhedral family with the local coefficient formula. Its utility in another family depends on proving the original support implication, computing a complete closure description, and finding an exact retained field. The rank-six success does not establish that these tasks remain tractable in higher rank.

The odd scalar formula and finite-character defect are also reusable beyond this instance, subject to their precise lattice and metric hypotheses. The primitive $0, \pm 1$ row condition explains the six-pole defect bound; it cannot be transferred to arbitrary integer matrices without checking the corresponding character supports.

The proof does not establish a positive tableau formula, monotonicity, concavity, superadditivity, or a simple local rule for paying each negative edge. Such statements are different from coefficient positivity, and some proposed monotonicity-related shortcuts were independently refuted during the investigation. A smaller conceptual field or a more local correction would improve explanation and verification cost. Those are substantive further questions, rather than missing premises of the stated certificate once its complete checks pass.

9 Contributions and AI assistance

This work was directed by Chris Hadad and involved substantial mathematical and computational contributions from AI systems across continuing research environments. The contribution record distinguishes discovery, construction, audit, independent reimplementations, and integration. The realizability route originated in an Opus 5.5 mathematical consultation and was developed into the complete pruned rational field by a separate Opus 5.5 exploration. Fable orchestrated and audited that work. Four fresh Opus-routed verification seats added proof reading, sampled exact classification, a separate complete classification and domain recount, and bounded whole-count controls. Two of those verification reports explicitly identify Opus 5.5; exact versions of the other two are unresolved in the inspected records. The Codex campaign subsequently supplied an independent proof-level adoption and complete original-normal geometric and arithmetic checks. Earlier ChatGPT Pro investigations and Codex work supplied the refined normal-cycle argument, higher fields, scalar algorithms, domains, and retained numerical checkpoints. The accompanying provenance table records exact artifact/model evidence and unresolved attribution details. During publication strengthening, a further ChatGPT Pro investigation supplied the legal-realization converse, completed the positive closure calculus and identified the minimum-six obstruction to its earlier rules. The publication campaign independently reviewed the argument, reimplemented its finite checker and constructive interface, and replayed all certificate predicates and the complete smaller-seed exhaustion. Those contributions strengthen the geometry and its accessibility; they are not a new run of the original numerical positivity field.

A model name in an orchestration wrapper is not necessarily the model that produced the mathematical response. Likewise, rerunning a producer’s program is different from writing a second verifier. The process description retains these distinctions. It records a cumulative research process in which separate contexts proposed, challenged, repaired and reused ideas; it is not a controlled comparison of model abilities or evidence of their relative credit percentages. AI review is distinct

from external human mathematical review. Established mathematics and the named researchers retain their explicit attribution.

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